Lecture 1: Limits

Notes render as you write: maths shows typeset as soon as the cursor leaves it. Press OptCmdN to jot something down without leaving this note, and CmdV to paste a photo of the board.

Definition

The sequence (a_n) **converges** to $L \in \mathbb{R}$ if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} : n \geq N \implies |a_n - L| < \varepsilon$$

Examples

- $\lim_{n \to \infty} \frac{1}{n} = 0$
- $\lim_{x \to 0} \frac{\sin x}{x} = 1$, see Lecture 2 for the proof.

To do

- ☐ Exercise sheet 1, questions 1 to 4
- ☒ Read chapter 2 of the course book